

# Does Reward Density Set the LoRA Rank Needed to Match Full Fine-Tuning?

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## Problem

LoRA matches full fine-tuning (FullFT) on supervised tasks at <1% of the parameters. Does that hold under RL, where the only teacher is a scalar reward — and does the rank you need depend on how much that reward tells you?

- **Question:** does the LoRA rank needed to match FullFT scale with reward density (bits / episode)?
- **Hypothesis:** denser reward → more capacity → higher rank to match FullFT.

**Why it matters:** whether to size the adapter to the reward, and what RL fine-tuning actually changes in the weights.

## Background

### LoRA vs. FullFT

LoRA freezes  $W$  and learns a low-rank  $\Delta W = BA$  ( $r \ll d$ ); FullFT updates all 1.7B parameters. Rank  $r$  is the capacity knob.

### Prior work & the gap

LoRA  $\approx$  FullFT parity is established for supervised fine-tuning [1,5]; whether it holds under RL, and how it scales with reward density, is untested. PRMs [3,4] and on-policy distillation [5] are studied for accuracy — not for the adapter capacity they require.

### One algorithm, three rewards

Same GRPO update for every cell; only the reward module changes, giving a clean density axis: **outcome** (correctness,  $O(1)$ ), **process** (per-step PRM,  $O(S)$ ), **distillation** (per-token KL to an 8B teacher,  $O(N)$  bits/episode).

## Methods

**Student** Qwen3-1.7B-Base · **Task** MATH · **Algorithm** GRPO.

**Eval** held-out greedy pass@1 (math-verify, scored identically for every regime, so rows are comparable).

**Matrix** — 3 regimes  $\times$  {LoRA ranks 1, 4, 16, 64, 256 + FullFT}  $\times$  3 seeds = 54 runs:

| Regime           | Reward signal                            | Granularity   |
|------------------|------------------------------------------|---------------|
| Outcome (a)      | correct / incorrect                      | 1 / episode   |
| Process (b)      | PRM score per reasoning step             | $S$ / episode |
| Distillation (c) | $\log \pi_T - \log \pi_\theta$ per token | $N$ / episode |

Config:  $N=16$ , 1536 tok, 100 rollouts.  $\sim 2.8$  h/run →  $\sim 153$  H200 GPU-hours on Modal (+  $\sim 55$  A100-hr for the 8B distillation teacher).

## Results

### Capacity: LoRA rank vs. FullFT, by regime

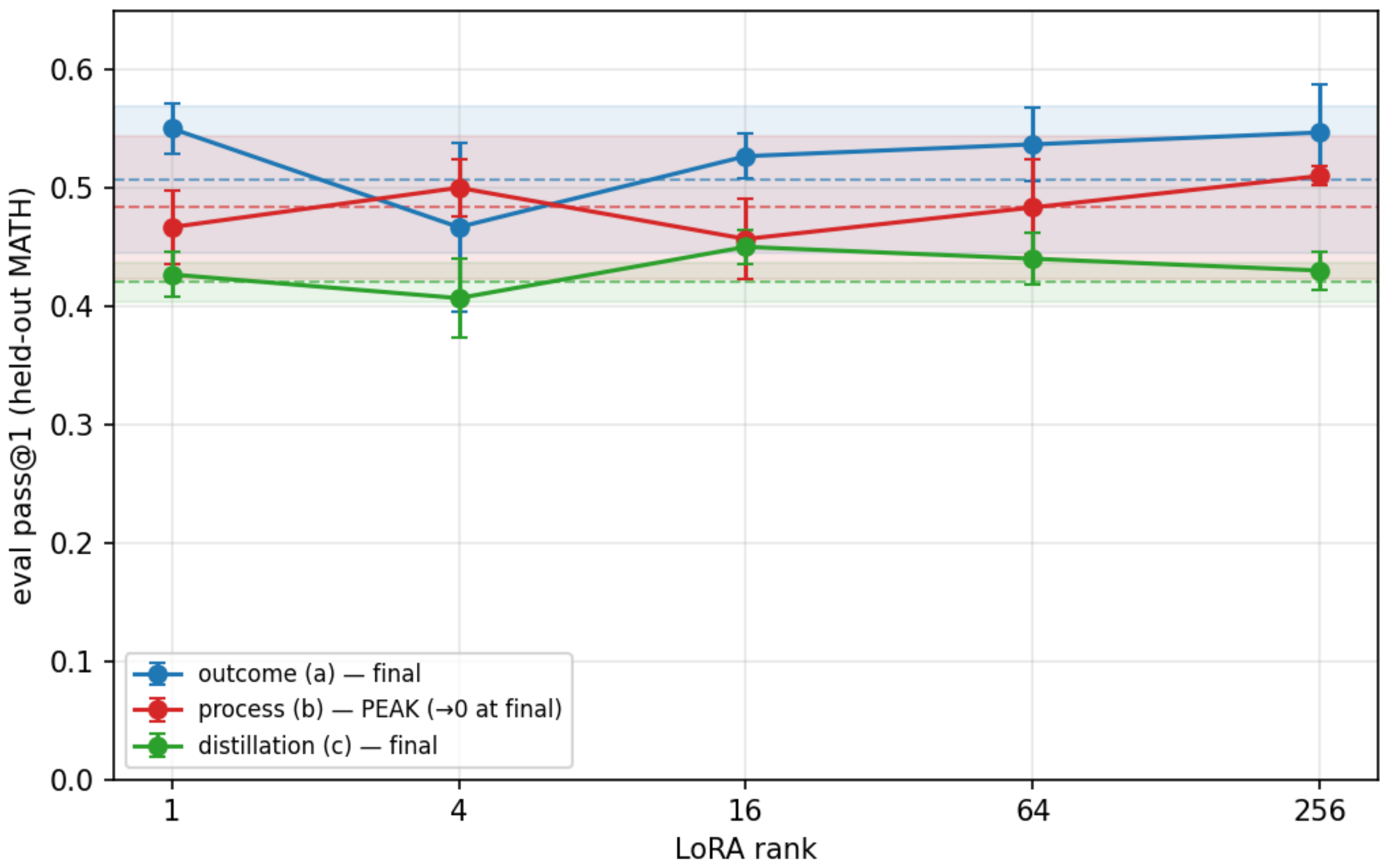


Figure 1. Held-out pass@1 vs. LoRA rank. Dashed line = FullFT mean, band =  $\pm 1$  sd. All three regimes are flat across rank and sit inside the FullFT band — rank 1  $\approx$  FullFT. (Process shown at its peak; it later collapses — see learning curves & Analysis.)

| pass@1         | r=1  | r=4  | r=16 | r=64 | r=256 | FullFT      |
|----------------|------|------|------|------|-------|-------------|
| Outcome        | 0.55 | 0.47 | 0.53 | 0.54 | 0.55  | <b>0.51</b> |
| Process (peak) | 0.47 | 0.50 | 0.46 | 0.48 | 0.51  | <b>0.48</b> |
| Distillation   | 0.43 | 0.41 | 0.45 | 0.44 | 0.43  | <b>0.42</b> |

3-seed means; per-regime sd  $\approx 0.02$ – $0.07$  (50-prompt eval).

### GRPO improves over the base model (0.38)

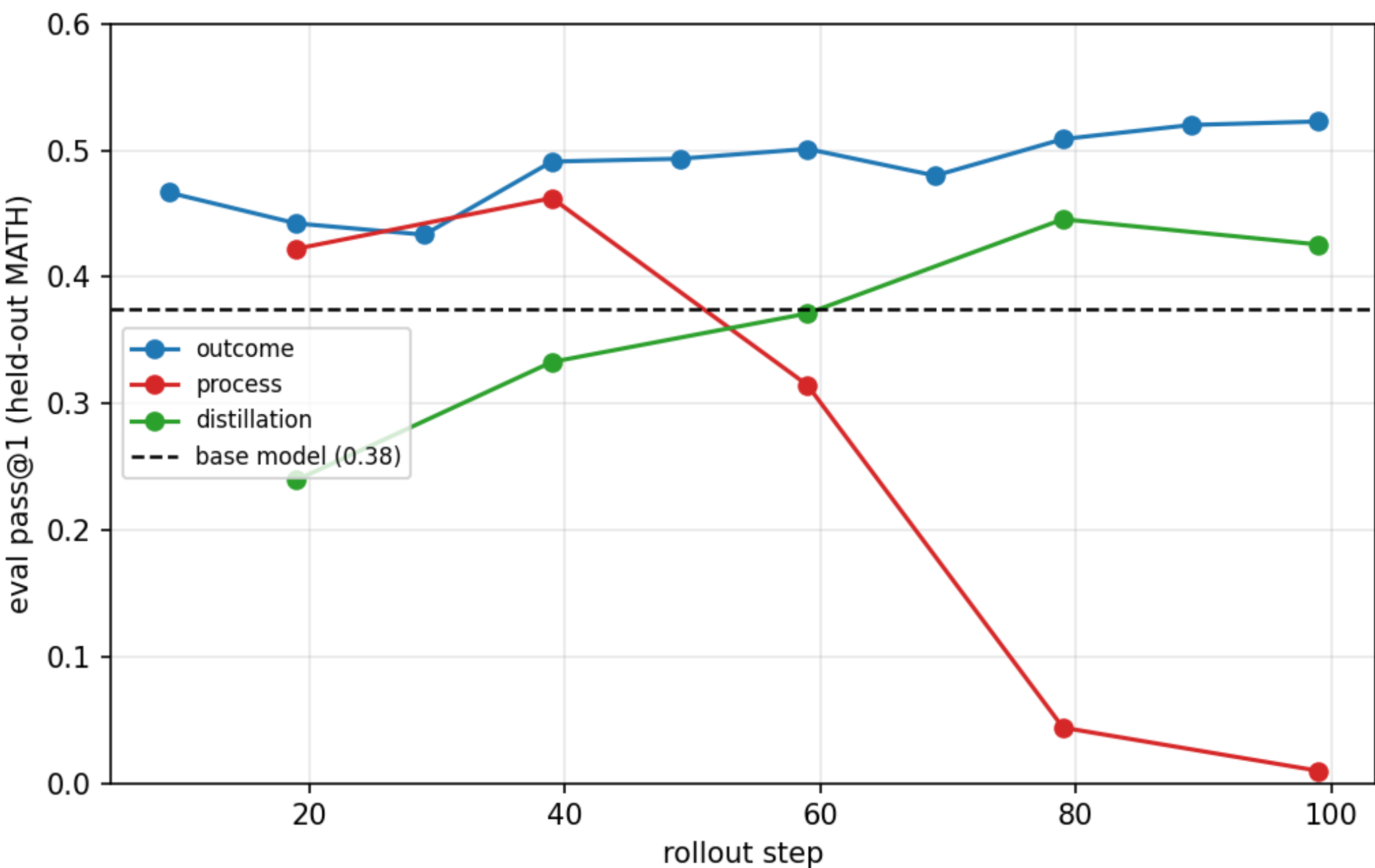


Figure 2. pass@1 over training; dashed = base (0.38). Outcome +0.15; distillation net +0.05 (dips below base on the teacher KL, then recovers above); process peaks then hacks below base.

The process curve (red) climbs to outcome-level accuracy then cliffs to  $\sim 0$ : the per-step reward rewards length, so completions hit the token cap ( $\text{frac\_truncated} \rightarrow 1$ ) and stop emitting answers (reward hacking; see Analysis).

## Analysis

### Finding 1: the capacity hypothesis fails — a clean null

- Across  $O(1) \rightarrow O(S) \rightarrow O(N)$  bits/episode, rank-1 LoRA matches FullFT (Fig. 1) — reward density does not raise the rank to match FullFT. Opposite of the hypothesis, consistent across the matrix.
- **Intuition:** RL on a capable base model is a low-rank nudge — a tiny adapter suffices regardless of reward detail.

### Finding 2: density's real cost is optimization stability — and it's reward structure, not density

- **Only process collapses.** The densest regime (distillation) is stable — so density itself isn't the problem.
- Process's step-sum reward (i) rewards length and (ii) makes the reverse-cumsum advantage spike (peak adv-std below); it collapses at every rank — even rank 1 — so it's the reward shape, not capacity.

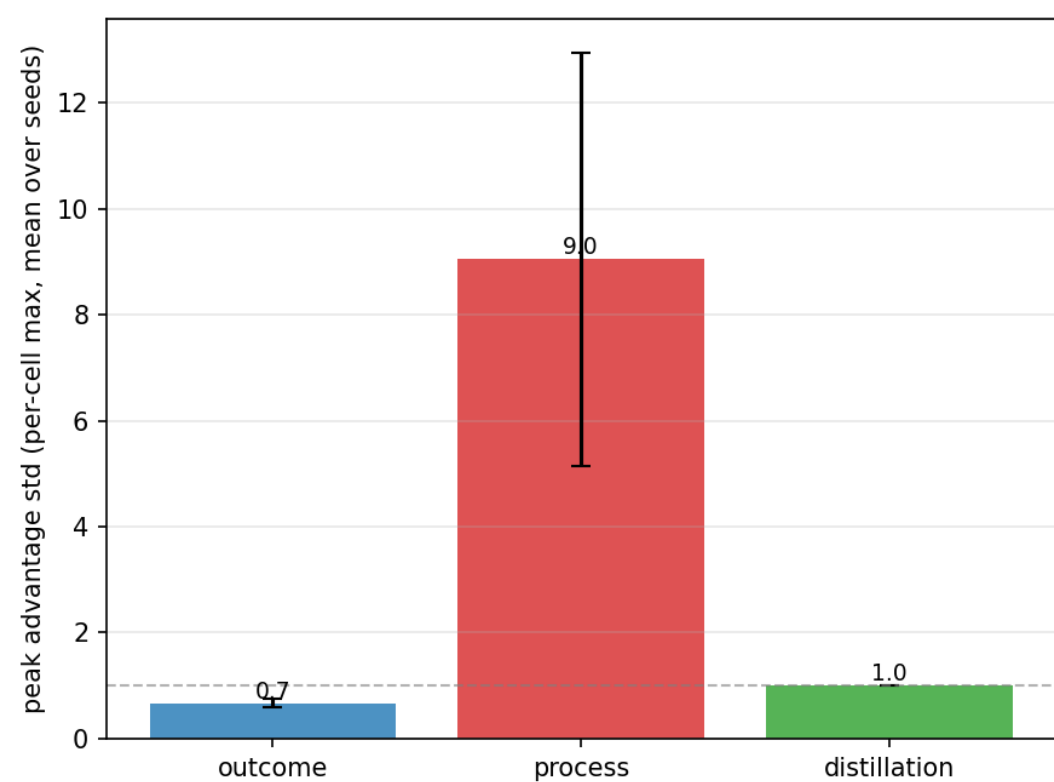


Figure 3. Peak advantage magnitude. Process's variable-step cumsum spikes ( $\sim 9$ ) vs. unit-scale outcome / distillation.

## Conclusions

- **Low rank suffices in every regime:** rank-1 LoRA matches FullFT for outcome, process (peak), and distillation — reward density does not drive adapter capacity.
- **Reward density trades off against optimization stability,** and the failure is the reward's shape (process's length-hackable step-sum), not its bit-count — the densest reward is the stablest.
- **Takeaway:** with a usable reward, spend effort on reward shaping, not adapter capacity.

**Limitations:** single task / 1.7B student; 50-prompt eval ( $\pm 0.07$ ); process “match” is best-checkpoint; the collapse is tied to the  $\backslash n$  segmentation — a better-shaped process reward may differ.

**Planned before the final report:** (1) re-run process with a length penalty / non-hackable segmentation — does a stable process reward show a rank effect?; (2) more rollouts to converge distillation.

## References

- [1] Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models.” ICLR, 2022.
- [2] Shao et al. “DeepSeekMath” (GRPO). arXiv, 2024.
- [3] Wang et al. “Math-Shepherd: Step-by-step Verification & RL.” ACL, 2024.
- [4] Lightman et al. “Let’s Verify Step by Step.” arXiv, 2023.
- [5] Thinking Machines Lab. “LoRA Without Regret” & “On-Policy Distillation.” 2025.